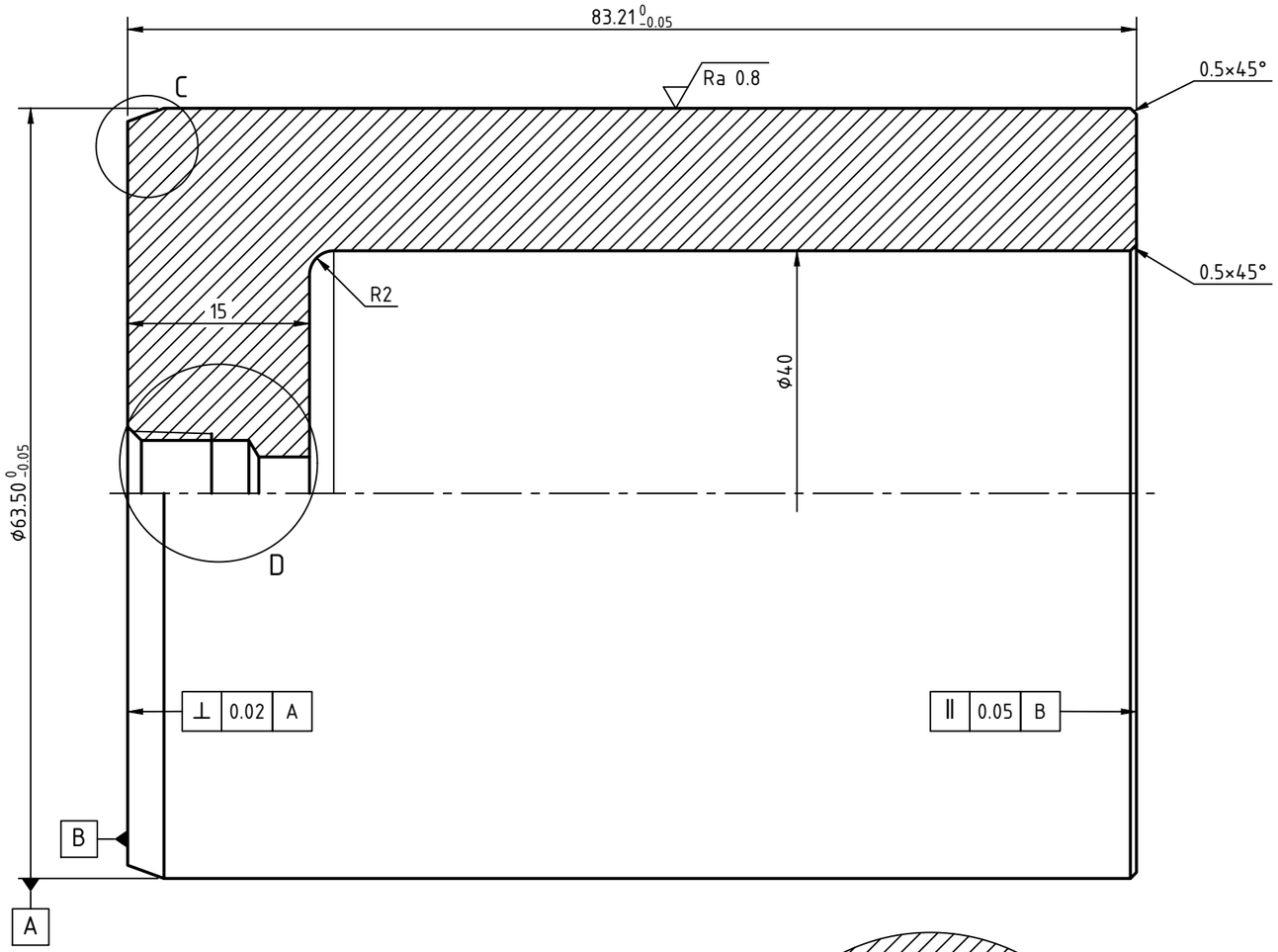
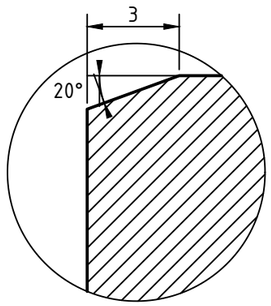
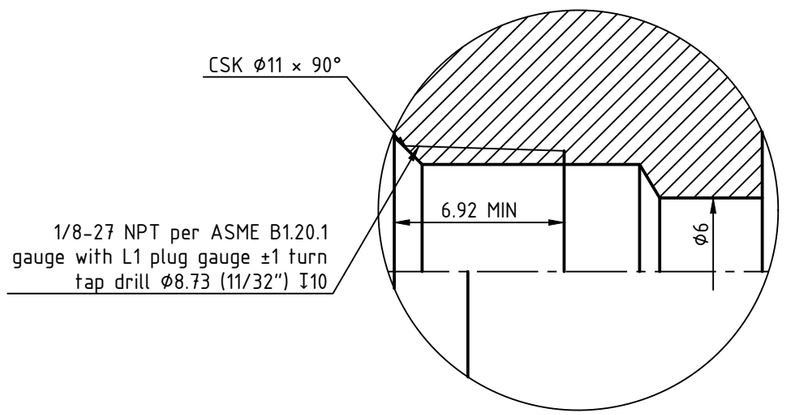




C (5:1)



D (4:1)

FILL / BLEED PORT (the motor's only port)

NOTES

1. Material: AISI 304 / 1.4301 austenitic stainless, annealed, round bar to ASTM A479. Material certificate EN 10204 3.1.
2. PROOF ONLY: never fired, never fitted for a firing. It takes the graphite insert SRM100-104's place in the hydrostatic proof SRM100-001, so the holder's lip, its bore gland and all five O-rings are proofed while the graphite stays dry.
3. Carries the insert's fits. OD ϕ63.50 0/-0.05: slip fit in the SRM100-102 bore ϕ63.58 +0.025/0, slides in by hand, aft end first; do not press. Length 83.21 0/-0.05 against the seat depth 83.21 +0.05/0: forward face flush to 0.10 below the holder face.
4. Face B bears on the holder's lip from ϕ54 out to ϕ61.32 (3.66 radial). Closed, the dummy puts 33 kN on the lip at 10.5 MPa, against the insert's 19 kN at MEOP 7 MPa.
5. OD Ra 0.8, free of axial scratches: the AS568-230 O-ring runs on it. Blend the lead-in into the OD, burr-free: the O-ring passes over it on assembly.
6. Port (detail D): the proof's fill and bleed port, the motor's only port (item 6 has none), on the axis of face B, reached through the holder's lip bore; the ϕ6 passage takes the ϕ4 dip tube (SRM100-001). 1/8-27 NPT per ASME B1.20.1: 90° countersink ϕ11, tap drill ϕ8.73 (11/32") I10, 118° point, then the passage into the cup. Full thread 6.92 min (L1 + L3): taper pipe tap with a 2 … 3 thread chamfer. Rig tee: Loctite 567 or equal on the male thread, first 2 threads clear; no PTFE tape.
7. Deburr, clean and degrease; passivate to ASTM A967. No chips or oil left in the cup, passage or port.
8. Inspect 100 %: OD at three axial positions, length, perpendicularity to A, port with a 1/8-27 NPT L1 plug gauge ±1 turn. Record the values against the serial.
9. Mark SRM100-120, serial and PROOF ONLY on the forward face, between ϕ44 and ϕ60, by laser or dot-peen, 2 mm characters, 0.05 deep max.
10. Extract as the insert: holder out, the insert push sleeve (tooling) through the lip bore onto face B. Never lever or knock.
11. USED ON: SRM100-001 hydrostatic proof assembly, item 17, in place of the insert SRM100-104 (item 3).

Ra 3.2 (✓)

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
AISI 304 / 1.4301, annealed round bar to ASTM A479	1414.1 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	2:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	PROOF DUMMY Hydrostatic-proof stand-in for insert SRM100-104 - 304 stainless

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-120	A	1 / 1